Validity Analysis: MASSIVE

Target context: Ethiopian Intercity Travelers — Amharic-Language Booking Bot

Overall risk: **HIGH**

Deployment Context

Use case: Amharic intercity-travel booking bots for Sky Bus, Selam Bus, and Ride covering routes, fares, and seat changes

Target population: Inter-regional commuters and travelers booking long-distance buses and ride-hail trips via smartphone apps

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology	2	Significant gaps	medium
Input Content ⚠	1	Serious concern	high
Input Form ✓	4	Minor gaps	medium
Output Ontology	2	Significant gaps	medium
Output Content ⚠	1	Serious concern	high
Output Form ✓	4	Minor gaps	high
Average	**2.3**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MASSIVE provides a strong modality and output-format match for an Amharic intent-classification + slot-filling booking bot, but exhibits severe content-level gaps for the Ethiopian intercity-travel deployment. The translation-from-English pipeline [Q2, Q4, Q106] guarantees that Amharic data lacks Ethiopian calendar dates, local city name variants, Birr fare conventions, and code-switching with Oromo or Tigrinya — all of which the user has confirmed appear in real user utterances. Operator-specific intents and slot types (operator_name, seat_class, route_closure_reason, surcharge_event, departure_terminal) are absent from the 60-intent / 55-slot taxonomy, which the paper itself describes as 'relatively simple' [Q108]. Annotator demographics for Amharic are undisclosed, raising convergent-validity concerns about whether labels reflect Ethiopian intercity-traveler judgments. Per-language data is small (~19.5k records, [Q103]) and Amharic — a unique-script, lower-resource language — is expected to fall in the lower performance tier per the script/pretraining-data correlation [Q87, Q98, Q100]. Strengths: text-only modality match, Ethiopic-script inclusion, output format directly compatible, and rich per-language breakdowns enabling expected-performance estimation.

ID	Evidence
Q2	"MASSIVE was created by tasking professional translators to localize the English-only SLURP dataset into 50 typologically diverse languages from 29 genera." (p.1)
Q4	"MASSIVE was created by localizing the SLURP NLU dataset (created only in English) in a parallel manner." (p.1)
Q87	"We compared the pretraining data quantities by language for XLM-R to its per-language task performance values, and in the zero shot setup, we found a Pearson correlation of 0.54 for exact match" (p.7)
Q98	"Discounting for artificial spacing effects, Germanic genera and Latin scripts performed the best overall (See Appendix E), which is unsurprising given the amount of pretraining data for those genera and scripts, as well as the quantity of Germanic and Latin-script languages in MASSIVE." (p.8)
Q100	"Icelandic was the lowest-performing Germanic language, likely due to a lack of pretraining data, as well as to its linguistic evolution away from the others due to isolated conditions." (p.8)
Q103	"Starting with the dataset, the per-language data quantities are relatively small at 19.5k total records and 11.5k records for training." (p.9)
Q106	"Third, the data were originally created through crowd-sourcing, not from a real virtual assistant, which introduces artificialities." (p.9)
Q108	"Fourth, our labeling schema is relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options." (p.9)

Practical Guidance

What This Benchmark Measures

MASSIVE provides a baseline for Amharic intent-classification and slot-filling on translated-from-English virtual-assistant utterances [Q2, Q5, Q73]. For the deployment, it can establish a lower-bound expectation for how XLM-R or mT5-class models handle generic booking-style intents in Amharic and provide per-language performance figures [Q126, Q127, Q128] useful for capacity planning. Output Form (score 4) and Input Form (score 4) are the strongest dimensions and indicate that the evaluation paradigm itself transfers cleanly.

Construct Depth

Construct depth is shallow for the deployment. MASSIVE measures translation-quality intent/slot accuracy on a generic taxonomy, not naturalistic Ethiopian intercity-booking utterances. Input Content (score 1) and Output Content (score 1) gaps mean the benchmark cannot probe Ethiopian-calendar slot extraction, Birr fare normalization, city-name-variant handling, code-switching, or L2-speaker phrasing — the phenomena most likely to drive real-world failure. Input Ontology (score 2) and Output Ontology (score 2) gaps mean operator-specific intents and slot types are entirely outside the evaluation surface.

ID	Evidence
Q2	"MASSIVE was created by tasking professional translators to localize the English-only SLURP dataset into 50 typologically diverse languages from 29 genera." (p.1)
Q5	"We also present modeling results on XLM-R and mT5, including exact match accuracy, intent classification accuracy, and slot-filling F1 score." (p.1)
Q73	"The decoder output is a sequence of labels (including the Other label) for all of the tokens followed by the intent." (p.7)
Q126	"Exact match accuracy by language for our three models using the full dataset and the zero-shot setup." (p.20)
Q127	"Table 8: Intent accuracy by language for our three models using the full dataset and the zero-shot setup." (p.21)
Q128	"Micro-averaged slot-filling F1 by language for our three models using the full dataset and the zero-shot setup." (p.22)